

AI Job Market Intelligence 2026

Data Analytics Project Report

Prepared for Data Analyst Internship Portfolio

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1. Executive Summary

This project analyzes 1,696 job postings to understand patterns in AI-related and traditional employment. The analysis covers job categories, companies, countries, employment types, salary distributions, search keywords, locations, and technical skills.

2. Introduction

The job market is changing as organizations adopt artificial intelligence, automation, and emerging technologies. This project uses job-posting data to examine how these changes appear in job demand, skills, salaries, employment types, and geographic patterns.

3. Project Objectives

- Analyze AI-related versus traditional job postings.
- Understand job-category distribution and company-level posting volume.
- Analyze job postings by country and location.
- Study employment and contract types.
- Analyze salary patterns and salary availability.
- Compare AI-related and traditional salary patterns within the available data.
- Identify frequently occurring job-search keywords.
- Extract and analyze technical skills mentioned in job descriptions.

4. Dataset & Tools

Dataset size: 1,696 job postings.

Key fields include title, company, location, category, contract type, contract time, salary range, description, posting date, currency, country, search keyword, and AI-related classification.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab/Jupyter Notebook.

5. Data Cleaning & Methodology

The notebook checks missing values, duplicates, and data types. No duplicate rows were found in the duplicate check. Missing categorical values were filled with 'Unknown'. Average salary was calculated from minimum and maximum salary where available. Exploratory analysis then examined AI classification,

categories, companies, countries, employment types, salaries, keywords, locations, and skills.

6. Key Findings

- The dataset contains 1,696 job postings.
- 1,021 postings (60.20%) are classified as AI-related and 675 (39.80%) as traditional.
- IT Jobs are the largest category with 695 postings, followed by Engineering Jobs (237), Admin Jobs (202), Customer Services Jobs (153), and Sales Jobs (133).
- The most frequent companies in the analyzed data include Bespoke Labs (112 postings), EE (111), and Maximus (71).
- The country distribution contains 750 postings from the United States, 746 from the United Kingdom, and 200 from India.
- For contract type, 1,264 records are marked Unknown, followed by permanent (289) and contract (143).
- The overall salary average in the dataset is about 82,288.50 and the median is 55,000. Salary values vary substantially across records and currencies.
- Among records with available salary values, the AI-related group has a mean salary of about 115,509.29 and median of 83,808.39, while the traditional group has a mean of about 37,217.92 and median of 32,540.91.
- London, UK is the most frequent location with 145 postings, followed by UK (54), India (44), Bangalore, Karnataka (38), and Belfast, Northern Ireland (23).
- The most frequent search keywords include artificial intelligence (450), automation (348), data entry (300), customer service (300), and machine learning (298).

7. Interpretation & Limitations

The findings describe this dataset and are not a complete representation of the global job market. Posting counts can depend on source coverage, search keywords, and geography. Salary comparisons should be made within the same currency and market rather than directly combining different currencies. Contract-type information is also incomplete for many records.

8. Business Recommendations

- Job seekers can use frequently occurring skills and keywords to guide skill development.
- Organizations can use job-market patterns to identify potential skill gaps.
- Recruiters can examine salary and skill patterns when designing hiring strategies.
- Educational programs can use observed demand patterns to align training with industry needs.

9. Conclusion

This project demonstrates how Python-based data analytics can be used to examine a job-market dataset and identify patterns across AI adoption, job categories, skills, salaries, employment types, companies, countries, and locations. It provides a portfolio example of data cleaning, exploratory analysis, visualization, and insight generation.

Project Files

- AI_Job_Market_Intelligence_2026.ipynb
- AI_Job_Market_Intelligence_2026.py
- ai_job_market_dataset.csv
- requirements.txt
- README.md